Complex Networks:
0/0 of Universal Connectivity

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The Removable Singularity Framework

Abstract

We show that complex networks exhibit a universal 0/0 removable singularity: the giant component appears at the percolation threshold. Scale-free networks have degree distribution P(k) ~ k^{-gamma} with gamma ~ 2-3, the SAME across Internet, social, biological, and financial networks. This is UNIVERSALITY for networks. The Watts-Strogatz small-world model has high clustering and short path lengths. The Barabasi-Albert model produces scale-free networks via preferential attachment. Networks are robust to random failure but fragile to targeted attacks. This connects to SOC, finance, consciousness, prebiotic chemistry, RMT, and epidemics.

1. Scale-Free Networks

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P(k) ~ k^{-gamma}  (gamma ~ 2-3)
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Scale-free networks have power-law degree distributions. Hubs (highly connected nodes) dominate. The Internet, social networks, biological networks, and financial networks all have gamma ~ 2-3. This is universality across all complex systems.

2. Three Network Models

- Erdos-Renyi (Ch.35): random graph, Poisson degree
- Watts-Strogatz: small-world, high clustering, short paths
- Barabasi-Albert: scale-free, preferential attachment

3. Small-World Property

Watts-Strogatz networks have high clustering (C >> C_random) and short path lengths (L ~ log(N)). This is "six degrees of separation" -- any two people are connected by a short chain of acquaintances.

4. Giant Component (0/0)

Below the percolation threshold: network is fragmented into small components. At the threshold: 0/0 -- the giant component appears. Above: the giant component dominates. This is the fundamental 0/0 of network connectivity.

5. Universality of Networks

ALL complex systems have the SAME network structure:

- Internet: gamma ~ 2.1
- Social networks: gamma ~ 2.5
- Biological networks: gamma ~ 2.3
- Financial networks: gamma ~ 2.8

This is universality for networks, just like Ising has universality for phase transitions.

6. Network Robustness

Scale-free networks are robust to random failure (hubs are rare) but fragile to targeted attack (remove hubs). The percolation threshold is p_c = 0 for scale-free networks -- they are always connected.

7. Connections

Network	Prior Chapter	Connection
Self-organization	Ch.41 SOC	Critical networks
Financial contagion	Ch.38 Finance	Network crashes
Neural connectivity	Ch.34 Consciousness	Brain network
Metabolic networks	Ch.35 Prebiotic	Life network
Adjacency eigenvalues	Ch.44 RMT	Spectral graph
Phase transitions	Ch.36 Ising	Network Ising

8. Conclusion

Complex networks exhibit universal 0/0 singularities. Scale-free networks have P(k) ~ k^{-gamma} with the SAME gamma across Internet, social, biological, and financial systems. The giant component appears at the percolation threshold (0/0). Networks are robust to failure but fragile to attack. This universality of networks connects to SOC, finance, consciousness, prebiotic chemistry, RMT, and epidemics.

References

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